

Introduction to Computer Programming

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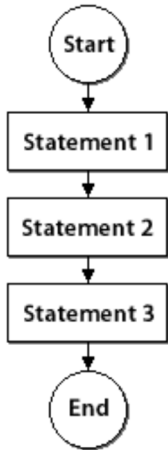
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Outline

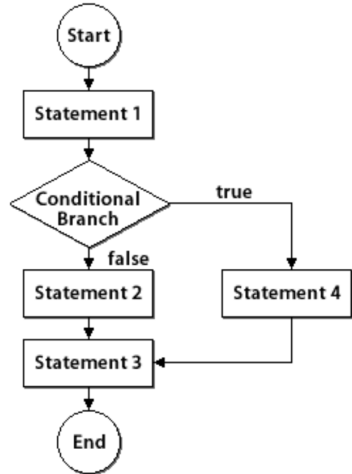
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Branching

A program **without branching**



A program **with branching**



Plain if

```
if <condition>:
    statement
    statement
    statement
    statement
```

if-else

```
if <condition>:
    statement
    ...
    statement
else:
    statement
    ...
    statement
```

if-else if-else

```
if <condition>:
    statement
    ...
    statement
elif <condition>:
    statement
    ...
    statement
else:
    statement
    ...
    statement
```

- <condition> yields True or False
- evaluate statements in that block if <condition> is True
- indentation is the key to delimit blocks
- place : at the end of <condition>

Examples

Examples if.

```
num = 3
if num>0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

```
num = -1
if num>0:
    print(num, "is a positive number.")
print("This is always printed. It's outside of the block.")
```

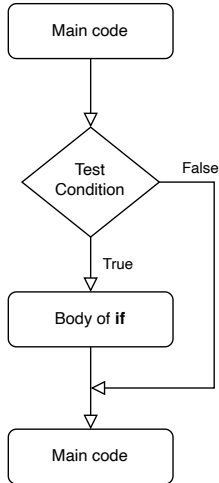
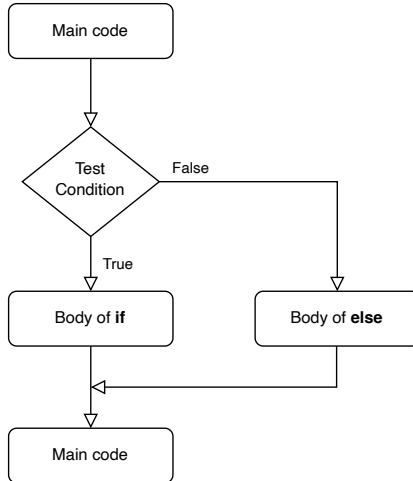


Figure 1: Logical flowchart of an if statement.



Examples if-elif-else.

```
num = 3.4
if num>0:
    print("Number is positive.")
elif num==0:
    print("Number is zero.")
else:
    print("Number is negative.")
```

What happens with

- `num=0`

or

- `num=-4.5`

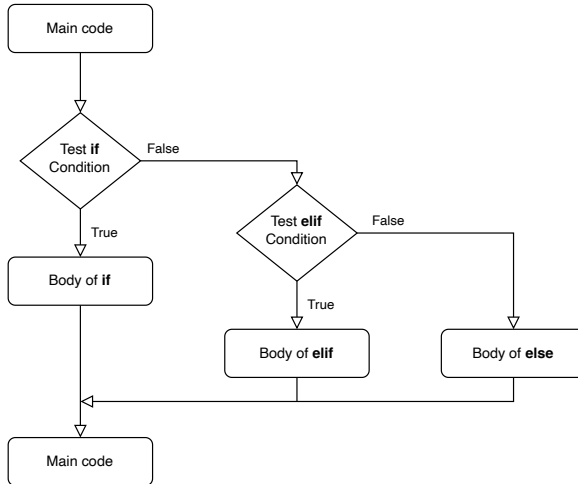


Figure 3: Logical flowchart of an `if-elif-else` statement.

Nested Statements

Branching statements `if`, `else` and `elif` can be nested arbitrarily.

```
num = input("Enter a number: ")
num = float(num)
if num >= 0:
    if num == 0:
        print("Number is zero")
    else:
        print("Number is positive")
else:
    print("Number is negative")
```

← recall: `"input()"` yields a string

Indentation

Recall

Indentation (tab or 4 spaces) delimits blocks of code in Python.

```
x = float(input("Enter a value for x: "))
y = float(input("Enter a value for y: "))
if x==y:
    print("y and x are equal")
    if y!=0:
        print("Finally, x/y is", x/y)
elif x>y:
    print("y is smaller")
else:
    print("x is smaller")
```

Exercises

- 1 Write a piece of code that checks whether a number is even. If so, print a message.

A possible solution follows:

```
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
```

- ② Write a piece of code that checks whether a number is even. Print a message about the outcome (odd or even).

A possible solution follows:

```
num = 42
if num % 2 == 0:
    print("Condition evaluates to True.")
    print("Number", num, "is even")
else:
    print("Condition evaluates to False.")
    print("Number", num, "is odd")
```

- ③ Write a piece of code that checks whether a number is odd or even. Assign a proper value (e.g., 'odd' or 'even') to a variable, depending on the outcome of the condition. Print the outcome at the end.

A possible solution follows:

```
num = 42
if num % 2 == 0:
    parity = "even"
else:
    parity = "odd"
print("Number", num, "is", parity)
```

- 4 Write a piece of code that checks whether a number is divisible by 2, 3, 5 and 7.

A possible solution follows:

```
num = 42
if num % 2 == 0:
    print("Number ", num, "is divisible by 2.")
if num % 3 == 0:
    print("Number ", num, "is divisible by 3.")
if num % 5 == 0:
    print("Number ", num, "is divisible by 5.")
if num % 7 == 0:
    print("Number ", num, "is divisible by 7.")
```


- 5 Write a piece of code that checks whether a number is odd or even. Print a message informing the user. If odd, check whether the number is divisible by 5. Print a message informing the user. If even, check whether the number is divisible by 3. Print a message informing the user.

A possible solution follows:

```
num = 15
if num % 2 == 0:
    print("Number", num, "is even.")
    if num % 3 == 0:
        print("Number", num, "is also divisible by 3.")
    else:
        print("Number", num, "is not divisible by 3.")
else:
    print("Number", num, "is odd.")
    if num % 5 == 0:
        print("Number", num, "is also divisible by 5.")
    else:
        print("Number", num, "is not divisible by 5.")
```

- 6 Write a piece of code that asks the user to insert the dog's age in human's years and calculates a dog's age in dog's years. For the first two years, a dog year is equal to 10.5 human years. After that, each dog year equals 4 human years.

A possible solution follows:

```
h_age = int(input("Input a dog's age in human years: "))
if h_age <= 0:
    print("Age must be a positive number.")
else:
    if h_age <= 2:
        d_age = h_age * 10.5
    else:
        d_age = 10.5 + 10.5 + (h_age - 2) * 4
    print("The dog's age in dog's years is:", d_age)
```

References

- Guttag, J. V. (2013). *Introduction to Computation and Programming using Python (Revised and Expanded Edition)*. MIT Press. [Chapter 2]
- Downey, A. (2015). *Think Python*. Green Tea Press ed. 2nd. [Sections 5.4–5.7]

